

Representative Structural Calculation Package

U-shaped school wing

Item	Value
Project	U-shaped school wing
Purpose	Representative elastic analysis and preliminary design trace
Generated	2026-06-27T00:00:00.000Z
Package version	m42-calculation-package
Source model	03-u-shaped-school

Nodes	Members	Loads	Combos	Max disp.	Max util.
40	57	174	28	63.66 mm	4.301

Model And Design Basis

Item	Value
Building type	school
Description	Three-story U-shaped education building around an open courtyard.
Occupancy	School
Bounds X/Y/Z	21.00 / 18.00 / 11.10
Analysis status	OK
Design status	OK
Package audit	OK

Load Case Trace

Case	Type	Loads	Fx	Fy	Fz
D	dead	27	0.000 kN	0.000 kN	-4309 kN
L	live	27	0.000 kN	0.000 kN	-2079 kN
WX	wind	30	89.91 kN	0.000 kN	0.000 kN
WY	wind	30	0.000 kN	104.89 kN	0.000 kN
EX	seismic	30	352.51 kN	0.000 kN	0.000 kN
EY	seismic	30	0.000 kN	352.51 kN	0.000 kN

Load Derivation

Story	Area	D total	L total	Wind X	Wind Y
1	378.00	1436 kN	831.60 kN	29.97 kN	34.97 kN
2	378.00	1436 kN	831.60 kN	29.97 kN	34.97 kN
3	378.00	1436 kN	415.80 kN	29.97 kN	34.96 kN

Load Derivation Formula Trace

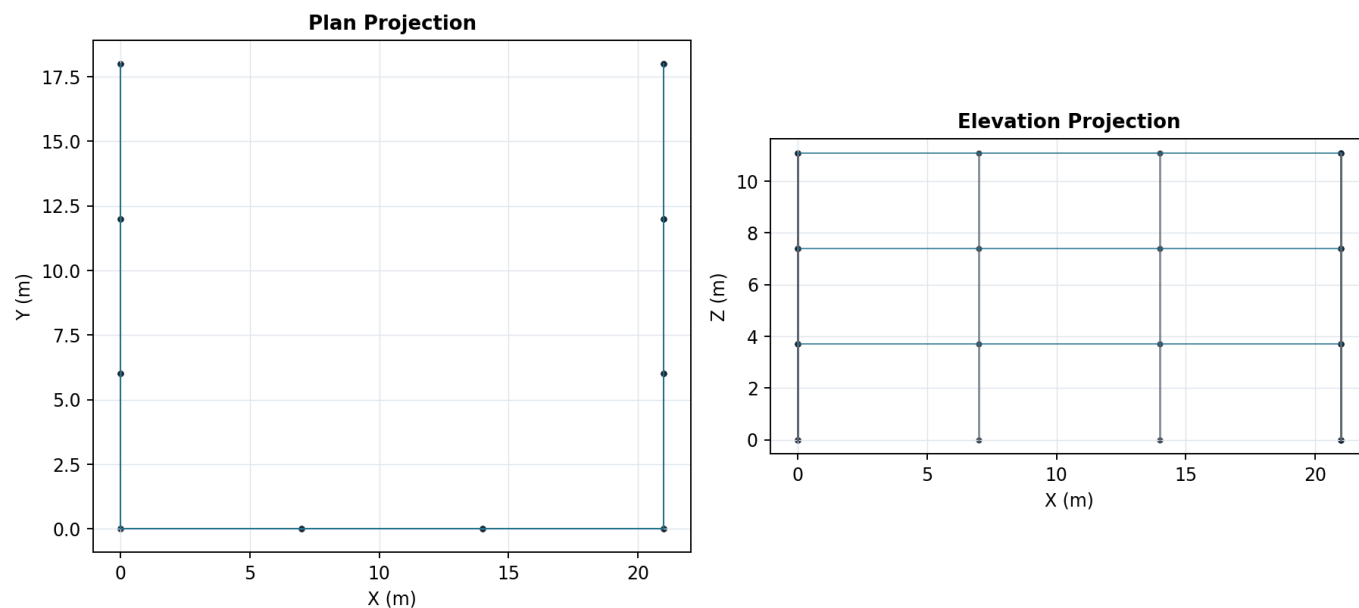
ID	Case	Formula	Result	Inputs
basis-occupancy	-	occupancy preset lookup	-	occupancy=school
basis-plan-area	-	$\max(\text{model width} * \text{model depth}, 1)$	378.00 m2	Bx=21.00 m, By=18.00 m
basis-roof-area	-	roofArea input or typical floor area	378.00 m2	roofArea=378.00 m2
D-ST1	D	$A * q_D$	1436 kN	A=378.00 m2, $q_D=3.800 \text{ kN/m}^2$
L-ST1	L	$A * q_L$	831.60 kN	A=378.00 m2, $q_L=2.200 \text{ kN/m}^2$
WX-ST1	WX	$p_{WX} * B_y * h$	29.97 kN	$p_{WX}=0.450 \text{ kN/m}^2$, By=18.00 m, h=3.700 m
WY-ST1	WY	$p_{WY} * B_x * h$	34.97 kN	$p_{WY}=0.450 \text{ kN/m}^2$, Bx=21.00 m, h=3.700 m
D-ST2	D	$A * q_D$	1436 kN	A=378.00 m2, $q_D=3.800 \text{ kN/m}^2$
L-ST2	L	$A * q_L$	831.60 kN	A=378.00 m2, $q_L=2.200 \text{ kN/m}^2$
WX-ST2	WX	$p_{WX} * B_y * h$	29.97 kN	$p_{WX}=0.450 \text{ kN/m}^2$, By=18.00 m, h=3.700 m
WY-ST2	WY	$p_{WY} * B_x * h$	34.97 kN	$p_{WY}=0.450 \text{ kN/m}^2$, Bx=21.00 m, h=3.700 m
D-ST3	D	$A * q_D$	1436 kN	A=378.00 m2, $q_D=3.800 \text{ kN/m}^2$
L-ST3	L	$A * q_L$	415.80 kN	A=378.00 m2, $q_L=1.100 \text{ kN/m}^2$
WX-ST3	WX	$p_{WX} * B_y * h$	29.97 kN	$p_{WX}=0.450 \text{ kN/m}^2$, By=18.00 m, h=3.700 m

KDS-Style Load Standard Audit

Symbol	Status	Mapped cases	Project input
D	mapped	D	Optional
L	mapped	L	Optional
Lr	missing	-	Optional
S	missing	-	Required
R	missing	-	Required
W	mapped	WX, WY	Required
E	mapped	EX, EY	Required
H	missing	-	Required
F	missing	-	Required
T	missing	-	Required

Preset	Status	Missing	Missing any	Generated
KDS-ST-03	blocked	-	Lr, S, R	0

U-shaped school wing



Combination Analysis Results

Combo	Status	Max disp.	Max util.	Residual
KDS-ST-01	OK	3.438 mm	0.403	0.000
KDS-ST-02	OK	4.248 mm	0.516	0.000
KDS-ST-04-WX-P	OK	13.69 mm	0.463	0.000
KDS-ST-04-WX-N	OK	13.69 mm	0.463	0.000
KDS-ST-04-WY-P	OK	5.882 mm	0.424	0.000
KDS-ST-04-WY-N	OK	5.431 mm	0.462	0.000
KDS-ST-05-EX-P	OK	63.66 mm	0.728	0.000
KDS-ST-05-EX-N	OK	63.66 mm	0.728	0.000
KDS-ST-05-EY-P	OK	20.03 mm	0.475	0.000
KDS-ST-05-EY-N	OK	19.51 mm	0.588	0.000
KDS-ST-06-WX-P	OK	13.40 mm	0.281	0.000
KDS-ST-06-WX-N	OK	13.40 mm	0.281	0.000
KDS-ST-06-WY-P	OK	5.227 mm	0.247	0.000
KDS-ST-06-WY-N	OK	4.937 mm	0.273	0.000
KDS-ST-07-EX-P	OK	63.38 mm	0.632	0.000
KDS-ST-07-EX-N	OK	63.38 mm	0.632	0.000

Serviceability Drift Review

Item	Value
Limit	H/200
Status	NG
Max drift	28.15 mm
Max drift ratio	0.00761
Governing	KDS-ST-05-EX-P / story 3

Combo	Story	Height	Drift	Ratio	D/L	Status
KDS-ST-01	1	3.700 m	0.083 mm	0.00002	0.004	OK
KDS-ST-01	2	3.700 m	0.218 mm	0.00006	0.012	OK
KDS-ST-01	3	3.700 m	0.364 mm	0.00010	0.020	OK
KDS-ST-02	1	3.700 m	0.109 mm	0.00003	0.006	OK
KDS-ST-02	2	3.700 m	0.292 mm	0.00008	0.016	OK
KDS-ST-02	3	3.700 m	0.448 mm	0.00012	0.024	OK
KDS-ST-04-WX-P	1	3.700 m	2.493 mm	0.00067	0.135	OK
KDS-ST-04-WX-P	2	3.700 m	5.260 mm	0.00142	0.284	OK
KDS-ST-04-WX-P	3	3.700 m	5.926 mm	0.00160	0.320	OK
KDS-ST-04-WX-N	1	3.700 m	2.493 mm	0.00067	0.135	OK
KDS-ST-04-WX-N	2	3.700 m	5.260 mm	0.00142	0.284	OK
KDS-ST-04-WX-N	3	3.700 m	5.926 mm	0.00160	0.320	OK
KDS-ST-04-WY-P	1	3.700 m	1.000 mm	0.00027	0.054	OK
KDS-ST-04-WY-P	2	3.700 m	1.778 mm	0.00048	0.096	OK
KDS-ST-04-WY-P	3	3.700 m	1.776 mm	0.00048	0.096	OK
KDS-ST-04-WY-N	1	3.700 m	1.066 mm	0.00029	0.058	OK
KDS-ST-04-WY-N	2	3.700 m	1.742 mm	0.00047	0.094	OK
KDS-ST-04-WY-N	3	3.700 m	1.377 mm	0.00037	0.074	OK
KDS-ST-05-EX-P	1	3.700 m	11.10 mm	0.00300	0.600	OK
KDS-ST-05-EX-P	2	3.700 m	24.41 mm	0.00660	1.320	NG

Governing Members

Rank	Member	Status	Util.	Check	Combo
1	M29	NG	4.301	steel-deflection	KDS-ST-05-EX-P
2	M30	NG	4.301	steel-deflection	KDS-ST-05-EX-N
3	M27	NG	3.369	steel-deflection	KDS-ST-05-EY-N
4	M28	NG	3.369	steel-deflection	KDS-ST-05-EY-N
5	M56	NG	2.652	steel-deflection	KDS-ST-05-EY-N
6	M57	NG	2.652	steel-deflection	KDS-ST-05-EY-N
7	M19	NG	2.399	steel-deflection	KDS-ST-05-EX-P
8	M20	NG	2.399	steel-deflection	KDS-ST-05-EX-N
9	M25	NG	2.084	steel-deflection	KDS-ST-05-EY-P
10	M26	NG	2.084	steel-deflection	KDS-ST-05-EY-P

11	M54	NG	2.077	steel-deflection	KDS-ST-05-EX-N
12	M55	NG	2.077	steel-deflection	KDS-ST-05-EX-P
13	M17	NG	1.860	steel-deflection	KDS-ST-05-EX-N
14	M18	NG	1.860	steel-deflection	KDS-ST-05-EX-P
15	M47	NG	1.480	steel-deflection	KDS-ST-05-EY-N
16	M48	NG	1.480	steel-deflection	KDS-ST-05-EY-N
17	M50	NG	1.285	steel-deflection	KDS-ST-05-EX-P
18	M53	NG	1.285	steel-deflection	KDS-ST-05-EX-N
19	M22	NG	1.224	steel-deflection	KDS-ST-05-EX-P
20	M23	NG	1.224	steel-deflection	KDS-ST-05-EX-N

Member Design Trace Matrix

Item	Value
Checked members	57
Unimplemented members	0
NG / WARN	24/17
Max utilization	4.301

Member	Type	Status	Util.	Formula	Action
M29	steel	NG	4.301	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M30	steel	NG	4.301	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M27	steel	NG	3.369	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M28	steel	NG	3.369	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M56	steel	NG	2.652	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M57	steel	NG	2.652	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M19	steel	NG	2.399	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M20	steel	NG	2.399	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M25	steel	NG	2.084	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M26	steel	NG	2.084	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.

M54	steel	NG	2.077	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M55	steel	NG	2.077	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M17	steel	NG	1.860	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M18	steel	NG	1.860	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M47	steel	NG	1.480	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M48	steel	NG	1.480	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M50	steel	NG	1.285	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M53	steel	NG	1.285	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M22	steel	NG	1.224	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.
M23	steel	NG	1.224	8	Revise member for Deflection.; Review serviceability deflection limit and member stiffness.

Steel, Connection, Foundation Summary

Scope	Rows	Warnings/NG	Max ratio
Steel review	57	41	4.301
Connections	57	34	4.299
Foundations	10	34	1.000

Quality Audit

Audit item	Status
Load derivation attached	OK
Combination results available	OK
Member checks available	OK
RC schedule available	Not applicable
Steel schedule available	OK
Foundation review available	OK

Remaining Scope

- Project design basis: occupancy, importance factor, site class, exposure, wind/seismic procedure, and serviceability criteria.
- Load derivation: dead, live, wind, seismic, snow, rain, soil, temperature, and construction load calculation sheets.

- Full KDS equation trace for every generated load combination and project-specific exception.
- RC member detailing: longitudinal bars, stirrups, development length, lap splice, anchorage, and constructability checks.
- Steel member detailing: compactness, lateral torsional buckling, connection forces, base plates, and weld/bolt checks.
- Foundation design: soil bearing, pile/footing design, settlement, uplift, sliding, overturning, and reinforcement.
- Drawing import and vision-assisted modeling audit trail for future CAD/image/MGT workflows.
- Drawing/image/MGT import audit trail is not attached to this generated test set.